Scalability & Future Plan

Date	1 November 2026
Team ID	N/A – Individual Project
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Maximum Marks	1 Mark

Scalability & Future Plan

The current solution is a Tableau Public report connected to a CSV dataset. Its next stage can focus on improving data freshness, handling larger datasets, strengthening validation and extending the analytical views. The upgrades below are future proposals, not features already implemented.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	CSV-based source; no automated refresh pipeline is documented.	New observations may require manual dataset replacement or reconnection.	High
2	Dashboard performance has not been measured under larger data volumes or many simultaneous viewers.	Responsiveness at larger scale is unknown.	Medium
3	No formal forecasting model, user authentication layer or application backend is included.	The current solution is descriptive and uses Tableau Public sharing.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Published Tableau Public Story; viewer capacity and concurrency were not tested.	For larger audiences, optimize workbook design and consider an organization-managed Tableau deployment if access control and capacity management are required.
2	**Data Storage**	CSV file is the current data source.	For larger or frequently updated datasets, evaluate a managed database or cloud data warehouse and a governed refresh process.
3	**Performance**	No benchmarked load time or filter response time was recorded.	Reduce unnecessary fields, optimize calculations and extracts where supported; record baseline load and interaction times before/after changes.
4	**Security**	Published Tableau Public content is publicly viewable; no private user-login layer is part of this project.	Do not publish confidential data. If private access is needed, move to an authenticated platform with role-based permissions and controlled data access.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Verify dataset quality, document field definitions, units, nulls and duplicate checks.	0–1 month	More reliable comparisons and clearer documentation.
Phase 2	Create a repeatable data-update process and evaluate a database-backed source if data volume grows.	1–3 months	Fresher data and improved maintainability.
Phase 3	Optimize workbook performance and add carefully validated filters or comparison views.	3–4 months	Smoother exploration and deeper analysis.
Phase 4	Explore forecasting/scenario analysis and a private deployment if required.	4–6 months	More advanced insights and stronger access control.

Roadmap note: Timelines are proposed planning estimates and may change with dataset availability, project scope and platform access.